

JSS1 Physics Diagnostic Bank (40 Questions)

Scope: Primary 5/6 Foundation & Full JSS1 Cambridge/Nigerian Physics

Focus: Physical Science Thinking & JSS2 Readiness

Section 1: Force, Motion & Energy (Questions 1 - 15)

No.	Question & Options	Ans	Source & Diagnostic Focus
1	A push or a pull acting on an object is called a: A) Energy B) Force C) Speed D) Mass	B	[P5] Definition of force.
2	Which force pulls objects toward the center of the Earth? A) Friction B) Magnetism C) Gravity D) Electricity	C	[P5/6] Gravitational force.
3	A car travels 100 meters in 10 seconds. Its speed is: A) 1000 m/s B) 10 m/s C) 90 m/s D) 110 m/s	B	[JSS1] Basic calculation of Speed ($Speed = Distance / Time$).
4	Which force opposes motion between two surfaces in contact? A) Gravity B) Friction C) Upthrust D) Tension	B	[P6/JSS1] Friction and its effects.
5	Energy due to an object's position or height is called: A) Kinetic Energy B) Potential Energy C) Chemical Energy D) Sound Energy	B	[JSS1/Cambridge] Energy forms.
6	Which of the following is a non-renewable energy source? A) Solar B) Wind C) Coal D) Hydroelectric	C	[P6/JSS1] Energy resources and sustainability.
7	When you rub your hands together, mechanical energy changes to: A) Light energy B) Heat energy C) Sound energy D) Chemical energy	B	[P5] Energy transformation.
8	The ability to do work is the definition of: A) Power B) Force C) Energy D) Pressure	C	[P5/6] Energy basics.
9	Work is done when a force moves an object through a: A) Time B) Distance C) Vacuum D) Circle	B	[JSS1] Concept of Work.
10	Which is an example of a simple machine? A) Pulley B) Car C) Computer D) Clock	A	[P5] Simple machines.
11	A Seesaw is an example of which type of lever? A) First Class B) Second Class C) Third Class D) Fourth Class	A	[JSS1] Lever classes.
12	The force that keeps an object floating in water is: A) Gravity B) Upthrust C) Friction D) Weight	B	[JSS1/Cambridge] Buoyancy/Upthrust.
13	Weight is measured using a: A) Beam balance B) Spring balance C) Ruler D) Thermometer	B	[JSS1] Force measurement tools.
14	The unit of Force is the: A) Joule B) Watt C) Newton D) Kilogram	C	[JSS1] SI units in Physics.
15	Mass is a measure of the amount of _____ in an object. A) Weight B) Matter C) Space D) Gravity	B	[P6/JSS1] Mass vs Weight concept.

Section 2: Light, Sound & Heat (Questions 16 - 30)

No.	Question & Options	Ans	Source & Diagnostic Focus
16	Light travels in: A) Circles B) Curved lines C) Straight lines D) Random paths	C	[P5/6] Rectilinear propagation of light.
17	The bouncing of light off a surface is called: A) Refraction B) Reflection C) Absorption D) Dispersion	B	[P6/JSS1] Reflection.
18	A rainbow is formed due to the _____ of light. A) Reflection B) Disappearance C) Dispersion D) Shadowing	C	[JSS1/Cambridge] Light spectrum.
19	Objects that allow all light to pass through them are: A) Opaque B) Translucent C) Transparent D) Reflective	C	[P5] Properties of materials.
20	Sound is produced by: A) Light B) Vibrations C) Heat D) Electricity	B	[P5/6] Production of sound.
21	Sound travels fastest through: A) Air B) Water C) Solids (e.g., Steel) D) Vacuum	C	[JSS1] Mediums of sound travel.
22	A reflected sound wave is called an: A) Echo B) Noise C) Pitch D) Volume	A	[P6] Sound reflection.
23	The transfer of heat through a solid material is: A) Convection B) Radiation C) Conduction D) Evaporation	C	[JSS1/Cambridge] Heat transfer modes.
24	Heat from the Sun reaches Earth through: A) Conduction B) Radiation C) Convection D) Insulators	B	[P6/JSS1] Radiation.
25	Material that does not allow heat to flow through it easily is an: A) Conductor B) Insulator C) Radiator D) Absorber	B	[P5] Thermal insulators.
26	Dark surfaces are good _____ of heat. A) Reflectors B) Absorbers C) Insulators D) Creates	B	[JSS1] Thermal radiation properties.
27	A periscope uses mirrors to see over obstacles. This uses: A) Reflection B) Refraction C) Sound D) Heat	A	[JSS1] Application of light properties.
28	The pitch of a sound depends on its: A) Amplitude B) Frequency C) Loudness D) Distance	B	[Cambridge JSS1] Sound characteristics.
29	Bending of light as it passes from air to water is: A) Reflection B) Refraction C) Dispersion D) Diffraction	B	[JSS1] Refraction basics.
30	Temperature is a measure of the average _____ energy of particles. A) Potential B) Chemical C) Kinetic D) Nuclear	C	[JSS1/Cambridge] Link between heat and kinetic theory.

Section 3: Electricity, Magnetism & Space (Questions 31 - 40)

No.	Question & Options	Ans	Source & Diagnostic Focus
31	A path through which electricity flows is a: A) Circuit B) Magnet C) Battery D) Switch	A	[P5/6] Basic electricity.
32	Materials that allow electricity to flow easily are: A) Insulators B) Conductors C) Resistors D) Plastics	B	[P5] Electrical conductivity.
33	Which component is used to open or close a circuit? A) Battery B) Bulb C) Switch D) Wire	C	[P5] Circuit components.
34	Like poles of two magnets (e.g., North and North) will: A) Attract B) Repel C) Disappear D) Heat up	B	[P6] Laws of magnetism.
35	A temporary magnet made using electricity is an: A) Natural magnet B) Electromagnet C) Lodestone D) Alloy	B	[JSS1] Electromagnetism introduction.
36	The Earth spinning on its own axis causes: A) Seasons B) Day and Night C) Years D) Eclipses	B	[P5/6] Earth's rotation.
37	The movement of the Earth around the Sun is called: A) Rotation B) Revolution C) Spin D) Orbiting	B	[P6] Earth's revolution and seasons.
38	Which planet is known as the "Red Planet"? A) Venus B) Mars C) Jupiter D) Saturn	B	[P5] Solar system facts.
39	A group of stars that forms a pattern is a: A) Galaxy B) Comet C) Constellation D) Asteroid	C	[JSS1] Astronomy basics.
40	Final Diagnostic Note: High scores in Sections 1 & 2 indicate strong readiness for Engineering and Physical Science tracks in SS1.	-	[Final Note] Pathway readiness indicator.